

Exact Generating Polynomials for Finite First Passage

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Abstract

We encode the twenty frozen first-passage laws of C88 as exact integer generating polynomials. Every polynomial has all seventeen coefficients, support spectrum, and derivatives at one through order six. Stirling inversion recovers the C89 ordinary moments exactly. The certificate is finite combinatorics under the stated scope firewall.

1 Polynomial certificate

For target H_i let $N_i(t) = \#\{\pi : T_i(\pi) = t\}$ and

$$G_i(z) = \sum_{t=0}^{16} N_i(t) z^t, \quad P_i(z) = G_i(z)/16!.$$

Then $P_i^{(m)}(1) = \mathbb{E}[(T_i)_m]$. With $S(r, m)$ denoting a Stirling number of the second kind,

$$\mathbb{E}[T_i^r] = \sum_{m=0}^r S(r, m) P_i^{(m)}(1).$$

Theorem 1. *All 20 polynomials have normalized nonnegative integer coefficients and the identity above recovers every C89 raw moment of orders 0 through 6.*

Proof. The coefficients are the exact C88 first-passage counts. Differentiating a finite polynomial gives falling-factorial moments, and Stirling inversion is the standard finite change of basis from falling factorials to powers. $\square$

certified object	count
target polynomials	20
coefficient cells	340
support/degree/gcd rows	20
derivative orders per target	7
raw-moment recoveries	140

2 Scope

No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside/table-of-marks, or Hilbert–Pólya operator is claimed.